Data Workflow Summary: Titanic Survival Analysis

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GitHub Repository: https://github.com/sandipdey2/ai-programming-foundations-project

Overview

This report describes a complete data workflow applied to the Titanic dataset. The dataset is available at https://www.kaggle.com/c/titanic. The workflow loads the data, applies two cleaning functions, performs exploratory analysis, creates three visualizations, and interprets the results. The GitHub repository containing the notebook, data, and version history is https://github.com/sandipdey2/ai-programming-foundations-project.

Dataset Description

The Titanic dataset contains 891 passenger records and 12 original columns. Key variables include Survived (binary target), Pclass, Sex, Age, Fare, and Embarked. After cleaning, identifier columns and the mostly-missing Cabin column were removed. Numeric variables such as Age and Fare are continuous; Sex, Pclass, and Survived are categorical. The data are historical passenger records from the 1912 Titanic voyage and are appropriate for academic exploration.

Workflow Description

The workflow begins by loading the CSV with Pandas. Two cleaning functions are then applied. `handle_missing_values` imputes missing Age with the median, fills missing Embarked with the mode, and drops the Cabin column. `remove_irrelevant_columns` drops PassengerId, Name, and Ticket because they are identifiers or high-cardinality text fields that do not support analysis. An exploratory function then computes summary statistics and survival rates by class and sex. Three labeled visualizations

follow: a survival count plot, a survival rate by sex bar chart, and a survival rate by passenger class bar chart.

Key Decisions and Assumptions

Median imputation was chosen for Age to reduce the influence of outliers. Mode imputation was used for Embarked because it is categorical and has few missing values. Cabin was dropped rather than imputed because more than three-quarters of the values are missing. The derived feature function used in earlier drafts was replaced with a second cleaning function to meet the requirement for two cleaning operations. These choices follow standard practice for preparing tabular data for exploration (McKinney, 2017).

Results and Interpretation

Figure 1 shows that fewer passengers survived than perished. Figure 2 shows that female passengers had a substantially higher survival rate than male passengers. Figure 3 shows that first-class passengers had a higher survival rate than second- or third-class passengers. These patterns are consistent with the grouped means produced by the exploratory function. The results suggest that sex and class were strongly associated with survival in this historical dataset.

Responsible Practice

The dataset reflects historical recording practices and social structures of 1912. Survival differences by sex and class may partly reflect access to lifeboats and social norms rather than purely biological factors. Interpreting these patterns as causal would be inappropriate. The analysis is descriptive only and does not support claims about modern safety or equity.

Reproducibility

All dependencies are listed in `requirements.txt` (generated with `pip freeze`). The notebook uses relative paths and runs top-to-bottom. The GitHub repository (https://github.com/sandipdey2/ai-programming-foundations-project) contains `main` and `develop` branches with multiple commits that record progress through data loading, cleaning, visualization, and documentation.

Future Integration Reflections

If these data were used to train a machine learning model, the cleaning steps would remain, but additional feature engineering and train-test splitting would be required. Before training a neural network, one would check for class imbalance, scale numeric features, and verify that no data leakage occurred during imputation. Agentic automation could help with repeated cleaning checks or documentation generation, but it would be risky for deciding imputation strategies without human review of the underlying missingness mechanism.

Sources and Citations

McKinney, W. (2017). *Python for data analysis* (2nd ed.). O'Reilly Media.

Wilson, G., Bryan, J., Cranston, K., Kitzes, J., Nederbragt, L., & Teal, T. K. (2017). Good enough practices in scientific computing. *PLOS Computational Biology, 13*(6), e1005510. https://doi.org/10.1371/journal.pcbi.1005510